K-Space: An Axiomatic Framework for Observer-Relative Measurement Registration in Quantum Theory

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Abstract

We introduce K-space, a formal structure defined by eight axioms (K1–K8) representing the content of observer-relative measurement registrations in quantum theory. K-space is not a physical space; it is a bookkeeping structure for what an observer has registered from quantum measurements, organized as a temporally ordered chain of K-state tuples, each carrying a registered outcome, a self-certification marker, a timestamp, and a validity status. The axioms define: the carrier set with admission and injectivity conditions (K1); a discrete temporal order (K2); intrinsic self-certification (K3); default validity with a null-event guard (K4); cross-observer invalidation dynamics (K5); cross-registration authority (K6); process closure transitioning provisional to final validity (K7); and cross-space embedding preservation (K8). A postulate K9_E extends K-space to a probability assignment $P(o \mid k_i, K_{\text{ctx}}) = \mathrm{Tr}(E_o\rho)[1 - \beta f_\perp]/Z_E$, where $\beta \in [0, 1]$; setting $\beta = 0$ recovers the Born rule as a corollary. Three bridge theorems (T1, T9, T5) connect the axioms to operational constructions in multi-observer scenarios; the supporting Level-4 admissibility conditions, the incommensurability and EWF bridges (T2, T3), the colimit existence theorem (T4-H), and the associativity proof (T5) are developed self-containedly in the appendices. The framework is interpretively neutral and requires no prior familiarity with any particular interpretive or motivational context.

Keywords: quantum measurement; axiomatic framework; observer-relative states; measurement registration; quantum foundations; K-space.

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1 Introduction

A recurring difficulty in quantum foundations is specifying, in formal terms, what it means for a particular observer to *have* a measurement outcome. Standard quantum mechanics is silent on this: the Born rule assigns probabilities to outcomes, and the Hilbert space formalism evolves states, but neither provides a structure-theoretic account of the content of observer-relative measurement registrations. Approaches that center on the observer — Everett's relative-state formulation [1], Wigner's Friend gedankenexperiments [2], and extended multi-observer scenarios [3, 4, 5] — expose the need for a precise bookkeeping structure for "what one observer has registered and how that registration relates to what another has registered."

Existing observer-relative frameworks do not fully supply this structure. Quantum Bayesianism (QBism) [6] locates quantum states in an agent's degrees of belief but does not axiomatize a registration-content object or a validity lifecycle for that content. Relational quantum mechanics [7] makes all properties observer-relative but leaves the internal structure of "what the observer has" informal. Consistent histories [8] assigns probabilities to sequences of projections at the ensemble level, not to individual observer registration events. The many-worlds interpretation [1] identifies branches through decoherence but provides no formal bookkeeping for the specific content that one branch-observer has registered versus another. The Spekkens toy model [9] provides axiomatic epistemic states, but these track knowledge of preparation procedures over ontic states, not time-stamped registration events with a validity lifecycle.

This paper introduces K-space to fill exactly this gap. A K-space K_R for registering system R is a set of tuples $k = \langle M, o, \mathrm{cert}, t, V\rangle$ representing individual registration events, equipped with a temporal chain order (K2), intrinsic certification (K3), default validity with invalidation dynamics (K4–K6), and a process-closure mechanism (K7). K-space is not a Hilbert space, not a phase space, and not a probability space; it is a registration-logic structure whose primitive predicates (cert, V, $\perp$, Auth) are epistemological rather than physical.

We also introduce postulate K9_E, which generalizes the Born rule by inserting a structural correction factor $[1 - \beta f_\perp]$ derived from the K-side context. At $\beta = 0$, K9_E reduces to the Born rule by corollary. At $\beta \neq 0$, K9_E produces measurable deviations in extended Wigner's Friend experiments; the empirical determination of β is addressed in a companion paper [10].

Paper structure. Section 2 states and motivates the eight axioms K1–K8. Section 3 introduces K9_E and derives the Born rule as a corollary. Section 4 presents three bridge theorems (T1, T9, T5). Section 5 compares K-space with existing frameworks. Section 6 states three open problems. Section 7 concludes. Appendices A.1–E make the paper self-contained: T8 and the Born corollary (A.1–A.2), restricted φ_R existence (A.3), the Level-4 admissibility definitions (B), the supporting bridges T2/T3/AJVS (C), the colimit existence theorem T4-H (D), and the associativity proof T5 (E).

2 K-Space Axioms K1–K8

Notation. $\mathcal{M}_K$ denotes the set of measurement-registration act identifiers (tokens, not types: two events of the same act type at different times are distinct tokens). O is the outcome alphabet; $\emptyset \in O$ is reserved for null events (zero information transfer). T_R is the discrete registration-time index set of R. $\Delta I(k)$ denotes the *information-transfer delta* of registration event k: the number of distinct field-values contributed by k to K_R beyond what a null token would contribute; $\Delta I(k) = 0$ iff k delivers no new information to K_R. Two K-state tuples k_1, k_2 are *K-incommensurable*, written $k_1 \perp_K k_2$, iff: (i) they fall within the same comparison context C_K (a Level-4 structural input supplied by the experimental protocol; defined in Appendix B, see also §6.2), and (ii) $o(k_1) \neq o(k_2)$ (conflicting registered outcomes). Equivalently, K5(ii) fires: $k_2 \perp k_1$ within C_K. For K-spaces, $K_{R_1} \perp_K K_{R_2}$ holds whenever $\exists\, k_1 \in K_{R_1}$, $k_2 \in K_{R_2}$ such that $k_1 \perp_K k_2$ (the generic K5-conflict trigger). This outcome-conflict condition is *sufficient* but not necessary: the complete admissibility-level characterization—under which $\perp_K$ may also arise from a closure lock (K7) with no outcome conflict—is Theorem T2 (Appendix C), which identifies $K_{R_1} \perp_K K_{R_2}$ with the non-existence of an admissible K_{joint}.

Scope Note

Axioms K1–K8 define K-space as a formal object. They do *not* derive quantum mechanics from K-space, nor establish the existence of a full structure-preserving map $\varphi\colon K \to \mathcal{B}(\mathcal{H})$.

Partial result. A restricted map $\varphi_R\colon K_R \to \mathcal{P}(\mathcal{H}) \cup \{0\}$ has been shown to exist for the $N = 2$ extended Wigner's Friend case (Class C theorem, 2026-06-01; Appendix A.3). This partial result does not imply the full φ-conjecture.

Open conjecture. The full map $\varphi\colon K \to \mathcal{B}(\mathcal{H})$ remains an open problem (§6.1).

Axiom 2.1 (Carrier Set — K1)**.** *The K-side registration space of a registering system R is a set K_R whose elements are K-state tuples:*

$$K_R = \left\{\, k \;\middle|\; k = \langle M,\ o,\ \mathrm{cert},\ t,\ V\rangle \right\}$$

with $M \in \mathcal{M}_K$, $o \in O \cup \{\emptyset\}$, $\mathrm{cert} \in \{0,1\}$, $t \in T_R$, $V \in \{0,1\}$.

Admission rule: $k \in K_R \Rightarrow \mathrm{cert}(k) = 1$ *(only self-certified events are admitted to* K_R*).*
Injectivity: $t \restriction K_R$ *is injective:* $\forall k_1, k_2 \in K_R,\ t(k_1) = t(k_2) \Rightarrow k_1 = k_2$*.*
Claim class: C (conjectural formal definition).

Remark 2.1. K_R records *what was registered*, not what physically exists. Elements k are not density matrices, not wave functions, and not probability values. The slot $o = \emptyset$ is reserved for null events (K4). K1 carries claim class C because the carrier-set definition is, as a whole, a conjectural formal claim about the existence of a well-formed registration space; K2–K8 carry class D as proposed structural operations acting on an already-admitted carrier. Among the five fields, M, o, cert, t are the immutable record of the event; V is the only *dynamic* field. The value V listed in the tuple is the validity status *at instantiation*; its subsequent value is the time-indexed status $V_{\mathrm{prov}}(k)$ (provisional, pre-closure) or $V_{\mathrm{final}}(k)$ (final, post-closure), governed by K4–K5 and K7. References to $V(k)$ in K5–K8 denote $V_{\mathrm{prov}}(k)$ before K7 closure and $V_{\mathrm{final}}(k)$ after.

Axiom 2.2 (Temporal Order — K2)**.** $(K_R, <_R)$ *is a strict total order (chain) where* $k_1 <_R k_2 \iff t(k_1) < t(k_2)$*. The order is* discrete*: for any consecutive pair* $k_i, k_{i+1} \in K_R$ *with no element strictly between them, no registration state exists in the open interval* $(t(k_i), t(k_{i+1}))$ *on the K-side.*
Claim class: D (proposed).

Remark 2.2. K2 does not claim physical time is discrete; only the K-side registration-time index within K_R is discrete. Physical time in Hilbert space remains continuous.

Axiom 2.3 (Self-Certification — K3)**.** $\sigma_R \colon \mathcal{M}_K \to \{0, 1\}$ *with*

$$\sigma_R(M) = 1 \iff M \text{ has occurred as a K-side registration event of } R,$$

determined intrinsically *within* K_R*. For all* $k \in K_R$*:* $\mathrm{cert}(k) = \sigma_R(M(k))$*.*
Observer-indexed independence: $\sigma_R(M)$ *is independent of* $\sigma_{R'}(M')$ *for any* $R' \neq R$*.*
Claim class: D.

Remark 2.3. $\sigma_R(M)$ certifies the *occurrence* of M as a K-side event. It does not certify that the physical outcome is "correct," is not consciousness, and is not a second-order measurement act.

Axiom 2.4 (Default Validity — K4)**.** *Define* $\mathrm{isNull}(k) := o(k) = \emptyset \wedge \Delta I(k) = 0$ *(zero-information-transfer event). For all* $k \in K_R$*:*
(a) $\neg\mathrm{isNull}(k) \Rightarrow V(k) = 1$ *upon instantiation,*
(b) $\mathrm{isNull}(k) \Rightarrow V(k) = 0$*.*
Claim class: D.

Remark 2.4. $V(k) = 1$ is the default K-side validity for non-null events: it means this registration is treated as valid within K_R until contradicted (K5). It does not mean the physical outcome is correct. Validity is provisional until process closure (K7).

Axiom 2.5 (Invalidation — K5)**.** *K5 fires only when a comparison context C_K exists (i.e., when joint registration is structurally demanded). $V(k_1) \to 0$ iff $\exists\, k_2$ in the relevant K-space such that:*

(i) $k_1 <_R k_2$ *(K2: k_2 is temporally later),*

(ii) $k_2 \perp k_1$ *within shared C_K* *(registered contradiction),*

(iii) $\mathrm{Auth}(k_2 \to k_1, C_K) = 1$ *(K6: cross-registration authority).*

K5 does not fire *when no C_K exists. Before process closure (K7), $V(k_1) = V_{\text{prov}}(k_1)$ is provisional and in principle reversible (if k_2 is itself invalidated before closure). After K7 closure, $V_{\text{final}}(k_1) = 0$ is absolute and irreversible.*

Claim class: D.

K5_prospective [EXTENSION — required for K9_E only]

For probability assignment via K9_E (§3), K5 admits a *prospective evaluation mode* on hypothetical tuples $k_o^* = \langle M^*, o, 1, t^*, 1\rangle$ representing a registration that *would* be instantiated if outcome o were realized.

K5 fires prospectively on k_o^* iff requires_K_joint $= 1$ and $\exists\, k_{\text{prev}} \in K_{\text{joint}}$ such that: (i) $k_{\text{prev}} <_{\text{joint}} k_o^*$, (ii) $k_o^* \perp k_{\text{prev}}$ within C_K, (iii) $\mathrm{Auth}(k_o^* \to k_{\text{prev}}, C_K) = 1$. Conditions (i)–(iii) are identical to K5; only the evaluation target differs. Prospective firing does *not* modify any V in K_R; it contributes only to $f_\perp(o)$ in K9_E (Theorem T8, Appendix A.1).

Axiom 2.6 (Cross-Registration Authority — K6)**.** $\mathrm{Auth}(k_2 \to k_1, C_K) = 1$ *iff all of:*

(a) C_K*-sphere*$(k_1) = C_K$*-sphere*(k_2) *(same comparison context),*

(b) $V(k_2) = 1$ *(k_2 not invalidated at check time),*

(c) $k_1 \in \mathrm{scope}(D_{\text{joint}})$ *(k_1's claim is within the joint-validity demand).*

Authority is non-hierarchical and non-transitive across distinct contexts. Two observers in the same C_K may hold mutual authority when both are valid.

Claim class: D.

Axiom 2.7 (Registration Process Closure — K7)**.** *R closes at $t_{\text{close}}(K_R)$ iff all pending joint-validity demands involving K_R are resolved. At closure, for all $k \in K_R$: $V_{\text{prov}}(k) \to V_{\text{final}}(k)$. Post-closure properties:*

(a) No new k can be instantiated in K_R,

(b) K5 irreversibility is absolute ($V_{\text{final}}(k) = 0$ is permanent),

(c) No new joint-validity demand D_{joint} involving K_R can be raised,

(d) K_{joint} involving K_R is final.

Claim class: D.

Axiom 2.8 (Cross-Space Embedding Preservation — K8)**.** *For any embedding $i\colon K_R \to K_X$ (where K_X is a joint or cross-space structure):*

(i) $V_X(i(k)) = V_R(k)$ *at embedding time* t_{embed} *for all* $k \in K_R$ *(validity snapshot preserved),*

(ii) All five tuple fields $(M, o, \text{cert}, t, V)$ *are preserved exactly under* i*,*

(iii) After embedding, $V_X(i(k))$ *evolves by K4–K7 rules within* K_X*; K8 does not immunize against future K5 invalidation in* K_X*.*

K8 is independent of K4: K4 governs validity at native instantiation; K8 governs validity-preservation at cross-space transfer.

Claim class: D.

Table 1 summarizes the layer-1 axiom cluster.

Table 1: K-space axioms K1–K8: roles, field coverage, and claim class. Class C = structurally testable conjecture; Class D = proposed. Neither modifies standard QM. "Lvl-4" indicates a conditional semantic dependency on Level-4 structural definitions (comparison context C_K, joint-validity demand D_{joint}, predicate requires_K_joint).

Axiom	**Name**	**Fields**	**Class**	**Lvl-4 dep.**
K1	Carrier set	M, o, cert, t, V	C	None
K2	Temporal order	t	D	None
K3	Self-cert.	cert	D	None
K4	Default validity	V (default)	D	None
K5	Invalidation	V (transition)	D	C_K
K6	Authority	V (auth. cond.)	D	C_K, scope(D_j)
K7	Closure	V (closure)	D	req.$_{KJ}$
K8	Embedding pres.	all five	D	None

3 K9_E: Probability Postulate

Axioms K1–K8 define K-space as a structural object but do not assign probabilities. Postulate K9_E supplies the probability assignment.

Definition 3.1 (K-context set)**.** For registering system R_i performing experiment Exp, the *contextual K-state set* is

$$K_{\text{ctx}}(k_i, \text{Exp}) = \Big\{ \varphi_{ij}(k_j) \in K_{\text{joint}} \;\Big|\; k_j \in K_{R_j}, \\ R_j \in \text{Obs}(\text{Exp}, R_i),\; k_j \text{ temporally compatible with } k_i \Big\} \quad (1)$$

where φ_{ij} is the K8-constrained canonical embedding constructed in Theorem T9 (§4) below; its existence and uniqueness are established there without additional assumption (beyond K1–K8 + T1). The (finite, R_i-relative) *observer set* is

$$\mathrm{Obs}(\mathrm{Exp}, R_i) := \{\, R_j \neq R_i : \text{requires\_K\_joint}(R_i, R_j) = 1 \text{ and } R_j \text{ participates in Exp} \,\};$$

this membership criterion (definition D_obs) replaces the earlier informal "other observer" clause and makes K_{ctx} fully formal (see Appendix B for the underlying Level-4 predicates). When $\mathrm{Obs}(\mathrm{Exp}, R_i) = \emptyset$ one has $K_{\mathrm{ctx}} = \emptyset$ and K9_E reduces to the Born rule (cf. the empty-context convention in Def. 3.2).

Definition 3.2 (Perpendicularity fraction)**.**

$$f_\perp(o, K_{\mathrm{ctx}}) = \frac{\left|\{k_j \in K_{\mathrm{ctx}} : k_o^* \perp_K k_j\}\right|}{|K_{\mathrm{ctx}}|} \tag{2}$$

where $k_o^* = \langle M^*, o, 1, t^*, 1\rangle$ is the hypothetical outcome-o tuple (cf. the K5_prospective box, §2.5). Since all $k_j \in K_{\mathrm{ctx}}$ share the comparison context, $k_o^* \perp_K k_j \iff o(k_j) \neq o$; thus $f_\perp(o, \cdot)$ counts the fraction of context registrations whose outcome *conflicts* with the candidate o, and is therefore genuinely o-dependent — the property that makes the $[1 - \beta f_\perp]$ factor survive normalization (cf. alternative A2 in Theorem T8). Equivalently by Theorem T8 (Appendix A.1):

$$f_\perp(o, K_{\mathrm{ctx}}) = \mathbb{E}\big[\mathbf{1}(\text{K5\_prospective fires on } k_o^* \text{ vs } k_j \in K_{\mathrm{ctx}})\big].$$

Empty-context convention: when $|K_{\mathrm{ctx}}| = 0$ (no other registering system satisfies requires_K_joint), set $f_\perp(o, K_{\mathrm{ctx}}) := 0$. By Eq. (3) this collapses K9_E to the Born rule, consistent with Corollary 3.0.1: an isolated observer incurs no K-side suppression.

Postulate 3.1 (K9_E Probability Postulate)**.**

$$\boxed{P(o \mid k_i, K_{\mathrm{ctx}}) = \frac{\mathrm{Tr}(E_o\rho)\,[1 - \beta\, f_\perp(o, K_{\mathrm{ctx}})]}{Z_E}} \tag{3}$$

where E_o is the POVM element for outcome o, ρ is the quantum state, $\beta \in [0,1]$ is a free parameter, and $Z_E = \sum_{o'} \mathrm{Tr}(E_{o'}\rho)\,[1 - \beta\, f_\perp(o', K_{\mathrm{ctx}})]$ ensures $\sum_o P(o \mid k_i, K_{\mathrm{ctx}}) = 1$.

Well-posedness ($Z_E > 0$). $Z_E > 0$ holds whenever there exists an outcome o' with $\mathrm{Tr}(E_{o'}\rho) > 0$ and $\beta\, f_\perp(o', K_{\mathrm{ctx}}) < 1$ (in particular for any $\beta < 1$, since $f_\perp \leq 1$ and $\sum_{o'} \mathrm{Tr}(E_{o'}\rho) = 1$). The sole degenerate case $Z_E = 0$ requires $\beta = 1$ together with $f_\perp(o', K_{\mathrm{ctx}}) = 1$ for every outcome o' carrying nonzero Born weight; in this degenerate boundary configuration K9_E is left undefined and the assignment defaults to the Born rule by the convention of Corollary 3.0.1.

Corollary 3.0.1 (Born Rule Recovery)**.** *At $\beta = 0$, K9_E recovers the Born rule exactly: $P(o \mid k_i, K_{\mathrm{ctx}})\big|_{\beta=0} = \mathrm{Tr}(E_o\rho)$.*

Proof. At $\beta = 0$: $Z_E = \sum_{o'} \mathrm{Tr}(E_{o'}\rho) = \mathrm{Tr}(\mathbf{1}\rho) = 1$ by POVM completeness. Therefore $P(o) = \mathrm{Tr}(E_o\rho)$. $\square$

Remark 3.1. K9_E is a generalization of the Born rule parameterized by β. At $\beta = 0$, K-space is present but produces no deviation from standard QM predictions. The empirical determination of whether $\beta \neq 0$ in nature is the scope of a companion paper [10], which proposes a modified Bong et al. protocol sensitive to K-side suppression effects.

4 Bridge Theorems

We present three bridge theorems connecting K1–K8 to operational constructions in multi-observer scenarios. The labels T1, T9, T5 follow the framework's internal numbering; the supporting results (Level-4 admissibility conditions, the incommensurability theorem T2, the EWF bridge T3, and the colimit existence theorem T4-H) are developed self-containedly in Appendices B–E. The three theorems are presented here in operational dependency order (T1 $\to$ T9 $\to$ T5) rather than index order.

4.1 T1: $\mathrm{K}_{\mathrm{joint}}$ Construction

Theorem 4.1 (Joint K-Space Construction — T1)**.** *Given registering systems A, B with* requires_K_joint$(A, B) = 1$ *via a shared joint-validity demand D_{joint}, there exists a candidate joint K-space $K_{\mathrm{joint}}(A, B)$ as the categorical colimit of the embedding diagram (K_A, K_B) with K1–K8-preserving morphisms $i_A\colon K_A \to K_{\mathrm{joint}}$ and $i_B\colon K_B \to K_{\mathrm{joint}}$.*

The combined order is the transitive closure

$$<_{\mathrm{joint}} = \big(i_A(<_A) \cup i_B(<_B) \cup \text{cross-rel}\big)^+,$$

where cross-rel *encodes cross-structure temporal relations from the shared laboratory history. $(K_{\mathrm{joint}}, <_{\mathrm{joint}})$ is a partial order; restricted to each $i_X(K_X)$ it is a chain (from K2).*

Remark 4.1. T1 is a *composition theorem*, not a pure K1–K8 derivation. It combines K1–K8 with external Level-4 inputs (cross-structure temporal relations from laboratory protocol). K1–K8 alone do not supply these relations. Existence of a candidate K_{joint} does not guarantee admissibility; the full admissibility conditions AdmJoint (i)–(v) are given in Appendix B.

4.2 T9: $\mathrm{K}_{\mathrm{ctx}}$ Construction

Theorem 4.2 ($\mathrm{K}_{\mathrm{ctx}}$ as Theorem — T9)**.** *When* requires_K_joint$(R_i, R_j) = 1$*, define $\varphi_{ij}\colon K_{R_j} \to K_{\mathrm{joint}}$ by $\varphi_{ij}(k_j) = i_j(k_j)$, where i_j is the K8-constrained canonical embedding* *from T1. Then φ_{ij} is the unique morphism preserving all five tuple fields, and K_{ctx} (Eq. (1)) is a theorem construction from K1–K8 + T1 with no additional assumption.*

Proof (four lemmas). **L1 (Existence).** requires_K_joint$(R_i, R_j) = 1 \Rightarrow$ T1 constructs K_{joint} with canonical embedding i_j. Define $\varphi_{ij} = i_j$; existence follows.
L2 (Uniqueness). Let $\psi\colon K_{R_j} \to K_{\text{joint}}$ satisfy K8 field preservation. For each k_j, $\psi(k_j)$ must have the same five fields as k_j (K8). By K1 t-injectivity, K_{joint} contains exactly one element with those fields: $i_j(k_j)$. Hence $\psi = \varphi_{ij}$; the morphism is structurally forced.
L3 (Field sufficiency). K8 preserves M, o, cert, t, V, which are exactly the fields required by K5_prospective (outcome o, temporal t, authority V, admission cert) and by K6 (validity V).
L4 (Exhaustion). Alternative channels are eliminated: (A) direct cross-space comparison without K_{joint} is undefined (no C_K); (B) ρ-side correlation is a category error (K-side requires tuple fields); (C) non-K8-compliant embeddings violate K8; (D) any other K8-compliant embedding equals φ_{ij} by L2. □

Remark 4.2 (Historical note: elimination of assumption [A-E1]). Earlier drafts of T9 required an explicit assumption [A-E1]: "a T3-morphism $K_{R_j} \to K_{\text{joint}}$ exists." Lemmas L1–L3 above show this assumption is redundant: the K8-constrained canonical embedding φ_{ij} exists (L1), is unique (L2), and supplies all fields required by K5_prospective and K6 (L3). Assumption [A-E1] is therefore eliminated; no assumption beyond K1–K8 + T1 is required. The supporting bridge theorems T2–T3 (Appendix C) record the general morphism and incommensurability theory from which [A-E1] was originally drawn.

4.3 T5: Associativity (Conditional)

Theorem 4.3 (K_{joint} Associativity — T5, conditional)**.** *Given admissible K-side spaces K_A, K_B, K_C:*

$$K_{\text{joint}}\big(K_{\text{joint}}(A, B),\, C\big) \;\cong\; K_{\text{joint}}(A, B, C)$$

as K1–K8-structured sets. This is conditional on: (C1) *the N-observer colimit existence theorem T4-H (Full Theorem, 4/4 steps verified 2026-05-28); and* (C2) *the global commutativity condition F7d for the $N = 3$ embedding diagram (confirmed via T4-H Step 3 content-basedness of K5).*

Remark 4.3 (T5: proof in Appendix E). A self-contained proof of T5 is given in Appendix E: the colimit existence theorem T4-H (Appendix D) supplies the universal property, and the content-basedness of K5 firing in the colimit (Appendix E, Lemma B2) establishes the global commutativity condition F7d for the $N = 3$ embedding diagram.

Remark 4.4 (Bridge_EWF application (T3, Appendix C)). In an Extended Wigner's Friend (EWF) configuration, Friend F registers definite outcome o_F and Wigner W registers the same laboratory in a coherent superposition. The EWF setup fixes the laboratory time order $t_F < t_W$ (so $k_F <_{\text{joint}} k_W$); the derivation presupposes this ordering and is not stated for $t_W < t_F$. Under candidate K_{joint}, K5 then forces a conflict: $k_W \perp k_F$ within C_K with $\text{Auth}(k_W \to k_F, C_K) = 1$, so $V(k_F) \to 0$ (at the admissibility level, at least one of $V(k_F), V(k_W)$ must drop), violating the admissibility condition. The firing *direction* is set by the laboratory time order, not by the axioms alone. This derivation additionally requires the Axiom of Joint Validity Semantics (AJVS — a named semantic postulate requiring that K_{joint} hosts original first-order K-side validity claims, not meta-descriptions of the form "within K_F, M_F registered $|h\rangle$"; see Appendix C for the full statement).

5 Relation to Existing Frameworks

Table 2 compares K-space with five existing frameworks.

Table 2: Structural comparison of K-space with observer-relative and epistemic quantum frameworks. CH: Consistent Histories. MWI: Many-Worlds. *Spekkens tracks epistemic states over ontic states, not time-stamped registration tuples with a validity lifecycle.

Property	**K-space**	**QBism**	**RQM**	**CH**	**MWI**	**Spekkens**
Observer-indexed	Yes	Yes	Yes	No	No (branch)	Yes
Formal axioms	K1–K8	No	Partial	Yes	Partial	Yes
Registration content	Yes	No	No	No	No	No*
Invalidation rule	K5	—	—	—	—	—
Closure property	K7	—	—	—	—	—
Born rule	K9_E (β=0)	Agent	Relat.	Std.	Derived	Analog

vs. QBism [6]. QBism centers on agent degrees of belief; K-space centers on formal registration content. K9_E is a structural postulate, not a subjective prior.

vs. Relational QM [7]. Both are observer-relative. K-space additionally axiomatizes the *content* of what is registered (K1–K4), an invalidation mechanism (K5–K6), and a closure property (K7). RQM leaves the registration structure informal.

vs. Consistent Histories [8]. CH assigns probabilities to projection sequences at the ensemble level. K-space operates at the level of individual observer registration events with a validity lifecycle.

vs. Many-Worlds [1]. Everett branching identifies branches through decoherence but provides no formal bookkeeping for the content of what one branch-observer has registered versus another. K-space supplies this via K1–K8 and the K_{joint} construction (T1).

vs. Spekkens toy model [9]. Spekkens tracks degrees of knowledge about preparation procedures (epistemic states over ontic states). K-space tracks registered *outcomes* as formal tuples $\langle M, o, \mathrm{cert}, t, V\rangle$ with a validity lifecycle (K4→K5→K7). These are orthogonal formalization targets: preparation-epistemic vs. outcome-registration-formal.

Non-claim. K-space is not presented as superior to these frameworks. It provides a minimal formal complement: axioms for registration content that the other frameworks leave implicit.

6 Open Problems

6.1 The full φ-map conjecture

Conjecture. Does there exist a full structure-preserving map $\varphi\colon K \to \mathcal{B}(\mathcal{H})$?

Partial result. A restricted map $\varphi_R\colon K_R \to \mathcal{P}(\mathcal{H}) \cup \{0\}$ has been shown to exist for the $N = 2$ EWF case (Class C theorem, 2026-06-01; Appendix A.3). Necessary conditions N_1–N_8 and N_T for the full map have been derived from K1–K8.

Why open. Extending the restricted existence result to the full K-space requires characterizing the image of φ in $\mathcal{B}(\mathcal{H})$ under all K1–K8 structure-preserving constraints — a question in the representation theory of registration-logic structures.

6.2 Observer set selection rule for K_{ctx}

Problem. K_{ctx} is formally constructed by T9 (Theorem 4.2), and the observer set $\mathrm{Obs}(\mathrm{Exp}, R_i)$ now has a formal membership criterion — requires_K_joint$(R_i, R_j) = 1$ together with participation in Exp (definition D_obs, stated with Eq. (1)). What remains open is not the *form* of Obs but its *extension*: the rule fixing *which* physical interactions set requires_K_joint $= 1$ is supplied by the experimental protocol (Level-4 D_{joint} scope), not by K1–K8 alone.

Why open. The predicate requires_K_joint encodes which physical interactions create mutual validity dependencies. A fully axiomatic treatment requires connecting the K-side registration layer to the experimental-design layer.

6.3 Full N-observer colimit chain

Problem. T4-H (N-observer colimit existence) is a Full Theorem (4/4 steps, 2026-05-28). T5 (composition associativity) is Class C conditional (on T4-H and global commutativity F7d). The full chain for $N > 3$ observers requires extending the F7d commutativity check to arbitrary observer diagrams.

Why open. Non-transitivity of K-side incommensurability ($K_A \perp_K K_B \wedge K_B \perp_K K_C \not\Rightarrow K_A \perp_K K_C$, by T4) means each pair in an N-observer configuration requires an independent check. For large N, the combinatorial structure of the $N(N-1)/2$ required checks may reveal structural constraints not visible at $N = 2, 3$.

7 Conclusion

We have introduced K-space, a formal structure defined by eight axioms (K1–K8) representing observer-relative measurement registrations in quantum theory. The axioms collectively define a carrier set with admission and injectivity conditions (K1), a discrete temporal chain order (K2), intrinsic self-certification (K3), default validity with a null-event guard (K4), cross-observer invalidation (K5), cross-registration authority (K6), process closure transitioning provisional to final validity (K7), and cross-space embedding preservation (K8). Three bridge theorems — T1 (K_{joint} construction), T9 (K_{ctx} as a derived theorem, eliminating a prior morphism assumption via four lemmas), and T5 (associativity, conditional) — connect K1–K8 to operational constructions in multi-observer scenarios.

Postulate K9_E generalizes the Born rule to $P(o \mid k_i, K_{\text{ctx}}) = \text{Tr}(E_o\rho)[1 - \beta f_\perp]/Z_E$, with $\beta = 0$ recovering the Born rule exactly (Corollary 3.0.1). The empirical determination of β is the subject of the companion paper [10].

Three open problems are identified with precise formulations: the full φ-map conjecture (with a partial restricted-existence result for the $N = 2$ EWF case), the observer set selection rule for K_{ctx}, and the full N-observer colimit chain. These problems define a research program.

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A Supplemental Proofs

A.1 Theorem T8: K5_prospective Frequency Bridge

Theorem A.1 (T8 — K5_prospective Frequency Bridge)**.** *Under K1–K8 binary primitives and a uniform context set* K_{ctx}*,*

$$f_\perp(o, K_{\text{ctx}}) = \mathbb{E}\big[\mathbf{1}(\text{K5\_prospective fires on } k_o^* \text{ vs } k_j \in K_{\text{ctx}})\big].$$

The fraction form (2) is the unique functional form consistent with binary K-side primitives.

Proof. For each $k_j \in K_{\text{ctx}}$, define indicator $I_j(o) = 1$ iff K5_prospective fires on k_o^* vs k_j (equivalently $k_o^* \perp_K k_j$, i.e. $o(k_j) \neq o$ within the shared C_K); else $I_j(o) = 0$. Then:

$$\frac{1}{|K_{\text{ctx}}|}\sum_j I_j(o) = \frac{|\{k_j : k_o^* \perp_K k_j\}|}{|K_{\text{ctx}}|} = f_\perp(o, K_{\text{ctx}}).$$

Uniformity of weight ($w_j = 1$ for all j) is structurally forced: all K-side primitives ($\perp$, Auth, V, cert, temporal order) are binary-valued in $\{0,1\}$ (K2–K6); no continuous contradiction-strength metric exists in K1–K8. Four natural alternative weightings are each independently eliminated by structural constraints: (A1) quantum-overlap weight is circular (uses Born-rule probabilities that K9_E is intended to modify); (A2) outcome-independent binary indicator collapses to $\delta P = 0$ under normalization; (A3) Auth-weighted form reduces to uniform because $\text{Auth} \in \{0, 1\}$ (K6); (A4) temporal-decay weight introduces a new free parameter beyond the β budget. The fraction form is the unique survivor. $\square$

A.2 Born Rule Corollary: Proof

Restated for completeness: at $\beta = 0$, $Z_E = \sum_{o'} \text{Tr}(E_{o'}\rho) = \text{Tr}(\mathbf{1}\rho) = 1$ by POVM completeness, so $P(o) = \text{Tr}(E_o\rho)$. $\square$

A.3 Restricted φ_R Existence (Summary)

Theorem A.2 (Restricted existence of φ_R)**.** *Let K_R be a K-space satisfying K1–K8, with outcome set $O = \{o_1, \ldots, o_m\}$ for measurement $M \in K_R$. Let $\mathcal{H}$ be a Hilbert space with* $\dim \mathcal{H} \geq |O|$ *and orthonormal basis $\{|o_i\rangle\}_{i=1}^m$. Define*

$$\varphi_R(k) \;:=\; \begin{cases} |o\rangle\langle o| & \text{if } V(k) = 1 \;\; (o = \text{outcome of } k), \\ 0_{\mathcal{B}(\mathcal{H})} & \text{if } V(k) = 0. \end{cases} \tag{4}$$

Then $\varphi_R \colon K_R \to \mathcal{P}(\mathcal{H}) \cup \{0\}$ is total and well-defined, and satisfies necessary conditions N_1–N_8 and N_T derived from K1–K8 and Theorems T1–T3 (verified in Table 3; T2 and T3 are stated in Appendix C). For N_T: when $K_{R_1} \perp_K K_{R_2}$, the two registrations stem from incompatible *measurements whose outcome projectors are non-orthogonal on the shared registration Hilbert space, so $[\iota(\varphi_R(k_1)), \varphi_R(k_2)] \neq 0$. (The orthonormal basis $\{|o_i\rangle\}$ indexes the outcomes of a* single *measurement M; outcomes of distinct incommensurable measurements are not mutually orthogonal, which is precisely what makes the commutator nonzero.)*

For $N \geq 3$ registering systems, a unique colimit extension $\varphi_{\text{colim}} \colon K_{\text{joint}}(R_1, \ldots, R_N) \to \mathcal{B}(\mathcal{H}_{\text{joint}})$ exists via the universal property of T4-H (Full Theorem, 2026-05-28) and satisfies conditions φ-N1 (uniqueness), φ-N2 (associativity, conditional on T5), and φ-N3 (pair commutator). The $N \geq 3$ extension is Class C conditional*: φ-N2 inherits the conditional status of T5 (§4, §6.3), whose proof is given in Appendix* E*. The $N = 2$ restricted existence result above is unconditional (Class C theorem).*

Proof. Construction (4) is explicit and unambiguous (K1 admission guarantees well-definedness; K4 dichotomy covers all $k \in K_R$). Every $V(k) = 1$ event of M carries a genuine outcome $o \in O$, so $|o\rangle$ is a basis vector; the reserved null slot $o = \emptyset$ has $V = 0$ by K4(b) and maps to $0_{\mathcal{B}(\mathcal{H})}$, so φ_R is total on K_R. Table 3 verifies each condition. For $N \geq 3$: the compatible family $\{\iota_i \circ \varphi_i\}$ satisfies T4-H Step 4 universality (Appendix D); the unique mediating morphism φ_{colim} is set equal to it (T5 associativity: Appendix E). $\square$

Table 3: Verification of necessary conditions N_1–N_8 and N_T for φ_R defined by Eq. (4).

Cond.	**Source**	**Verification**	**OK**
N_1 (well-def.)	K1	φ_R total; image closed under Lüders products	✓
N_2 (Lüders ord.)	K2	$t(k_1)<t(k_2) \Rightarrow P_{o_2}P_{o_1}P_{o_2}$ Lüders	✓
N_3 (cert-refl.)	K3	$P_o^2=P_o=P_o^\dagger$; $\mathrm{Im}(\varphi_R) \subseteq \mathrm{Proj} \cup \{0\}$	✓
N_4 (valid-pos.)	K4	$V=1 \Rightarrow \|o\rangle\langle o\| \geq 0$; isNull$\Rightarrow \varphi_R=0$	✓
N_5 (inval.-abs.)	K5	$V \to 0$ absorbing: $\varphi_R=0_{\mathcal{B}(\mathcal{H})}$	✓
N_6 (auth.-comp.)	K6	Auth$=1 \Rightarrow P_{o_2}P_{o_1} \neq 0$ (necessary); sufficiency: boundary	✓*
N_7 (closure-fin.)	K7	φ_{final} fixed at t_{close}; no post-closure change	✓
N_8 (embed.-nat.)	K8	$\varphi_X \circ i = \iota \circ \varphi_R$; $\iota(P_o) = P_o \otimes \mathbf{1}$	✓
N_T (K-incomm.)	T2/T3	$K_{R_1} \perp_K K_{R_2} \Rightarrow [\iota(\varphi_R(k_1)), \varphi_R(k_2)] \neq 0$ (non-orth. projectors)	✓

*Necessary direction holds; the converse (sufficiency, $P_{o_2}P_{o_1} \neq 0 \Rightarrow \text{Auth} = 1$) is the boundary of Theorem A (box below). N_T holds because incommensurable registrations map to non-orthogonal projectors on the shared registration Hilbert space (Appendix C, T2/T3).

Boundary of restricted existence

The map φ_R on $\mathcal{P}(\mathcal{H}) \cup \{0\}$ is proven. Two structural limits apply, with proof sketches below:

Theorem A (N_6 sufficiency boundary). The necessary direction $\mathrm{Auth}(k_2 \to k_1, C_K) = 1 \Rightarrow P_{o_2}P_{o_1} \neq 0$ holds; the converse is *not provable* from $\mathcal{B}(\mathcal{H})$ alone. Sketch (4 steps): (1) $\mathcal{B}(\mathcal{H})$ encodes only operator-algebraic data (projectors, products, commutators, spectra), not epistemic-sphere membership or authority scope; (2) by K6, Auth $= 1$ requires C_K-sphere co-membership (a) and $k_1 \in \mathrm{scope}(D_{\mathrm{joint}})$ (c), both K-side predicates with no operator-algebraic definition; (3) two projectors with $P_{o_2}P_{o_1} \neq 0$ can lie in *different* C_K-spheres (independent experiments sharing $\mathcal{H}$), which $\mathcal{B}(\mathcal{H})$ cannot distinguish from co-membership; (4) hence no $\mathcal{B}(\mathcal{H})$-only proof of the converse exists.

Theorem B (global commutativity boundary, $N > 2$). The pairwise conditions $K_{R_i} \perp_K K_{R_j} \Rightarrow [\iota_i(P_{o_i}), \iota_j(P_{o_j})] \neq 0$ hold, but global K_{joint} path-commutativity is *not* determined by them. Sketch (2 steps): (1) $K_{\mathrm{joint}}(R_1, \ldots, R_N) = \mathrm{colim}(D)$ is fixed by a global universal property (Appendix D, Step 4), and path-commutativity is a global categorical property, not decomposable into pairwise relations without the coherence supplied by T5 (Appendix E); (2) the $N(N-1)/2$ pairwise commutators encode only binary relations, so no pairwise $\mathcal{B}(\mathcal{H})$ data recovers global colimit coherence.

These are characterized limits, not defects, and reflect current understanding rather than permanent impossibility: an operator-algebraic encoding of C_K-sphere membership would resolve Theorem A, and a $\mathcal{B}(\mathcal{H})$-expressible global coherence condition would resolve Theorem B.

φ_R is a *correspondence map* with precisely characterized limits. $\mathrm{Im}(\varphi_R) = \{|o\rangle\langle o|\} \cup \{0\}$ is the tightest $\mathcal{B}(\mathcal{H})$ subset needed; φ_R does not map to density operators, observables, or any operator without a direct K-registration analogue.

B Level-4 Structural Inputs

Axioms K5–K7 and the bridge theorems reference four structural predicates supplied by the experimental-design layer ("Level 4"). They are collected here for self-containedness; none modifies K1–K8.

Shared validity demand D_{joint}. $D_{\mathrm{joint}}(A, B, \mathrm{Arch}) \in \{0, 1\}$ evaluates to 1 when the comparison architecture Arch (the experimental design) demands that A and B support one shared registration-validity claim. D_{joint} is a structural feature of Arch, not imposed by any single observer.

Joint-check predicate requires_K_joint**.** requires_K_joint$(A, B) = 1$ iff all of: (a) A and B are each valid or provisionally valid within their own K-side; (b) A, B are brought under a shared D_{joint}; (c) D_{joint} requires both to be assessed as parts of the same registration target, history, counterfactual claim, or validity claim; (d) the truth of D_{joint} cannot be evaluated while leaving A, B in fully independent K-sides; (e) preserving D_{joint} requires a K_{joint} in which A, B are jointly valid. It is 0 iff no shared D_{joint} is imposed, or D_{joint} can be evaluated without embedding A, B into one candidate K_{joint}.

Comparison context C_K. C_K is the minimal shared frame in which two registration acts are evaluated for compatibility. It requires: (a) both acts admitted into the same comparison domain; (b) both indexed to the same registration target; (c) comparison does not presuppose joint validity. C_K is strictly weaker than AdmJoint: it enables comparison without requiring that joint validity be preserved.

Admissible joint space AdmJoint**.** AdmJoint$(K_{\text{joint}}; A, B) = 1$ iff there exist embeddings $i_A : A \to K_{\text{joint}}$ and $i_B : B \to K_{\text{joint}}$ such that:

(i) the embeddings preserve act, outcome, certification, registration time/order, and validity;

(ii) self-certification remains intrinsic to each embedded act;

(iii) the six validity conditions below remain satisfied for each embedded structure;

(iv) no required registration-state update in K_{joint} invalidates either embedded structure while both are still claimed as jointly valid;

(v) K_{joint} does not import an external certifier as the source of self-certification.

The six *validity conditions* for a registered measurement are: physical occurrence; admission across the K-side boundary; process membership; self-certification ($\sigma_R(M) = 1$); default validity ($V = 1$); and non-invalidation (no later contradicting registration).

C Supporting Bridge Theorems: T2, T3, and AJVS

These results support T9 and the EWF application (Remark in §4). T2 and T3 are Class D; AJVS is a declared semantic postulate. None modifies K1–K8.

C.1 T2 — K-Incommensurability Derivation

Theorem C.1 ($\perp_K$ derivation — T2)**.** *For K-spaces K_A, K_B: $K_A \perp_K K_B$ iff* requires_K_joint$(A, B) = 1$ *and no admissible K_{joint} exists satisfying* AdmJoint *(i)–(v) (Appendix B) while preserving K4 (default validity) and K5 (no invalidation) for both embedded sides simultaneously.*

Derivation (K5-conflict trace). Under a candidate K_{joint}, suppose $\exists\, k_A \in i_A(K_A), k_B \in i_B(K_B)$ with $k_B \perp k_A$ within C_K and $\text{Auth}(k_B \to k_A, C_K) = 1$ (K6). Then K5 forces

$V_{\text{prov}}(k_A) \to 0$ or $V_{\text{prov}}(k_B) \to 0$, violating AdmJoint (iv) (no invalidation while both are jointly claimed valid). Hence no admissible K_{joint} exists, i.e. $K_A \perp_K K_B$. The K5 conflict is *sufficient*, not necessary: admissibility can also fail by a closure lock (K7: if K_A has already closed, no new act demanded by D_{joint} can be instantiated, so AdmJoint (i) fails with no K5 contradiction).

C.2 T3 — Extended Wigner's Friend Bridge

Theorem C.2 (Bridge_EWF — T3). *In an Extended Wigner's Friend configuration with laboratory time order $t_F < t_W$, where M_F registers a definite outcome o_F ($k_F = \langle M_F, o_F, 1, t_F, 1\rangle$) and M_W registers the same laboratory as a coherent superposition in which no definite o_F is preserved ($k_W = \langle M_W, o_W, 1, t_W, 1\rangle$): under a candidate K_{joint}, $k_W \perp k_F$ within C_K with* $\mathrm{Auth}(k_W \to k_F, C_K) = 1$, *so K5 forces* $V(k_F) \to 0$ *(the authority direction $k_W \to k_F$, fixed by the laboratory order $t_F < t_W$, makes k_F the invalidated side; at the admissibility level it suffices that at least one of $V(k_F), V(k_W)$ drop), violating* AdmJoint *(iv). Consequently $M_W \perp M_F$ at the act level.*

The derivation presupposes the EWF ordering $t_F < t_W$ (it is not stated for $t_W < t_F$) and is conditional on AJVS below.

C.3 AJVS — Axiom of Joint Validity Semantics

Postulate C.1 (AJVS). *A candidate K_{joint} satisfies $D_{\text{joint}}(A, B)$ iff it hosts the* original *K-side validity claims of both K_A and K_B as jointly evaluable first-order claims in a shared C_K. Hosting only meta-descriptions of the form "within K_A, M_A registered o_A" does* not *satisfy D_{joint}: relativizing registration contents to sub-context descriptions abandons D_{joint} rather than satisfying it.*

AJVS is a declared semantic postulate, separate from K1–K8. If AJVS is rejected, the T3 conclusion does not follow from K1–K8 alone; K1–K8 remain structurally valid, and only the semantic scope of D_{joint} changes.

D N-Observer Colimit Existence (T4-H)

Let $\mathcal{C}_K$ be the category whose objects are K-spaces (sets satisfying K1–K8) and whose morphisms are K1–K8-preserving embeddings.

Theorem D.1 (Colimit existence — T4-H). *Every finite diagram D in $\mathcal{C}_K$ has a colimit* $\mathrm{colim}(D)$ *in $\mathcal{C}_K$.*

Verification (4 steps).

S1 Category. $\mathcal{C}_K$ has identities, composition, and associativity of K1–K8-preserving embeddings.

S2 Construction. $K_{\text{colim}} = \left(\coprod_i K_i\right)/\sim$, with all five tuple fields well-defined: t assigned lexicographically and V as the embedding-time snapshot (K8); the order $<_{\text{colim}}$ is the transitive closure of the embedded orders plus cross-structure relations (T1-generalized).

S3 K1–K8 preservation. All eight axioms survive the quotient. Key lemma (T-PRES): K8-morphisms preserve t, so $t_{\text{colim}}([k,i]) := t_i(k)$ is representative-independent; K2 acyclicity follows, K5 cross-component $\perp$ is content-based, and the monotone V-dynamics are non-contradictory.

S4 Universal property. For any compatible cocone $\{f_i\}$, the map $u([k,i]) := f_i(k)$ is well-defined (diagram compatibility + T-PRES), K8-preserving, and unique.

Hence K_{colim} exists, satisfies K1–K8, and is universal. The construction is constructive for $N = 2$ (this is T1) and extends to all finite N.

E Associativity of K_{joint} (T5)

We prove T5 (§4) conditional on: (C1) T4-H (Appendix D); (C2) requires_K_joint = 1 with AdmJoint satisfied for each pairwise and triple construction; (C3) the global commutativity condition F7d, established below.

Theorem E.1 (Associativity — T5, conditional)**.** *Under* (C1)–(C3)*:* $K_{\text{joint}}\big(K_{\text{joint}}(A,B),C\big) \cong K_{\text{joint}}(A,B,C)$ *as K1–K8-structured sets.*

Lemma B1 (field preservation). Any K8-preserving morphism preserves all five tuple fields as equalities; in particular $t_{\text{colim}}([k,i]) = t_i(k)$ is path-independent (T-PRES).

Lemma B2 (content-based K5). For $[k,i],[k',j]$ in the colimit with $i \neq j$, K5 fires iff (i) $i \neq j$ (observer identity), (ii) $o([k,i]) \neq o([k',j])$ (outcome content), and (iii) $t_{\text{colim}}([k,i]), t_{\text{colim}}([k',j])$ are co-temporal. All three depend only on tuple content, hence are path-independent.

Theorem B3 (F7d holds). By B1–B2, the set of K5 events firing against any k is identical along the incremental and one-shot construction paths; therefore V is path-independent and the global commutativity condition F7d holds.

Lemmas A1–A4 (universal property). $K_{\text{joint}}(A,B)$ and the iterate $K_{\text{joint}}(K_{\text{joint}}(A,B),C)$ are objects of $\mathcal{C}_K$ (T4-H Step 3). The composed embeddings j_A, j_B, j_C form a valid cocone for the 3-diagram $D = \{K_A, K_B, K_C\}$ (B3 gives V-agreement), and $K_{\text{joint}}(K_{\text{joint}}(A,B),C)$ satisfies the universal property for D: the T1 universal property gives a unique u_{AB}, and the T4-H universal property (Appendix D, S4) then gives a unique mediating u, with uniqueness inherited from both.

Proof of T5. Both $K_{\text{joint}}(K_{\text{joint}}(A, B), C)$ (Lemmas A1–A4) and $K_{\text{joint}}(A, B, C)$ (T4 + T4-H) are colimits of the same diagram D. Two colimits of one diagram are canonically isomorphic, so the unique mediating morphisms compose to identities in both directions, and the isomorphism is K1–K8-preserving. Thus K_{joint} is associative up to K1–K8-preserving isomorphism. T5 asserts K-side structural isomorphism only — not ρ-side associativity, physical outcome composition, or any modification of standard QM. $\square$